
RICE: Reference-guided Internal Cross-lingual Erasure for Multilingual LLM Unlearning

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Abstract

Machine unlearning aims to remove targeted behaviors from a trained language model without retraining from scratch, but multilingual LLMs make this objective harder: information suppressed in one language may remain recoverable through another, while aggressive deletion can damage non-target utility. We present RICE, a hybrid fine-tuning objective for multilingual factual unlearning that combines reference-guided output suppression with internal representation erasure. RICE couples Negative Preference Optimization (NPO) with Representation Misdirection for Unlearning (RMU), which redirects selected forget-set hidden states toward fixed random directions. We evaluate RICE on a ten-language translated TOFU setup using Qwen3.5-2B and a 1% target-forgetting split. Importantly, unlearning is supervised only on English forget examples; the other nine languages are used only for evaluation. Despite this single-language supervision, RICE achieves a strong Truth Ratio Forget (*TR Forget*) score of 0.8552 while preserving a high downstream model utility of 0.3538, close to the retrained reference baseline (0.3645). Compared with standard negative preference and gradient-based deletion baselines, which suffer from weak unlearning transfer or high utility costs, RICE defines the Pareto-optimal trade-off frontier between utility preservation and cross-lingual factual erasure. These results suggest that compact late-layer representation misdirection can extend deletion pressure across translated factual prompts while preserving multilingual model utility.

1 Introduction

Large language models (LLMs) can memorize private, copyrighted, unsafe, or otherwise undesirable information during training. Removing such information by retraining from a filtered corpus is often impractical, motivating approximate machine unlearning: given a forget set and a retain set, the goal is to suppress target behavior while preserving unrelated model utility. For LLMs, this is difficult because target information can be represented through many prompts and surface forms, and because deletion pressure can easily spill over into broadly useful linguistic or factual capabilities [1, 2].

Multilingual models introduce an additional challenge. A factual association may be encoded in partially shared semantic representations while being expressed through language-specific tokens and scripts. Consequently, deleting a target in one language need not delete it in another, and the utility cost of deletion can vary across languages. Recent multilingual unlearning studies report weak or asymmetric cross-lingual transfer, language-dependent retention, and low-resource-language leakage

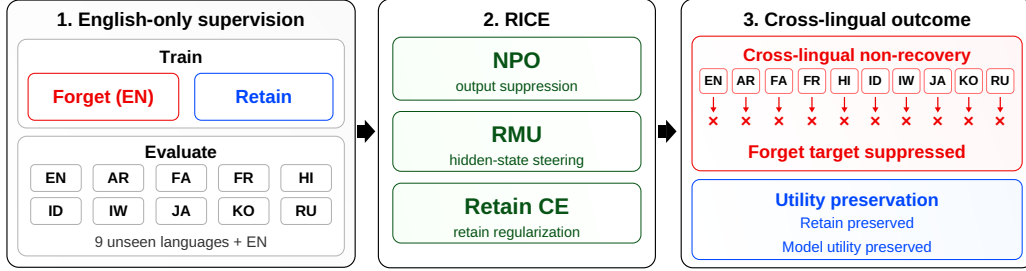


Figure 1: Overall pipeline of RICE. The model is trained with English-only forget supervision and retain examples, while evaluation is conducted across ten languages. RICE combines NPO for output-level suppression, RMU for hidden-state steering, and retain CE for utility preservation, enabling cross-lingual non-recovery of the forget target.

risks [3, 4, 5, 6]. These observations suggest that multilingual unlearning should be evaluated not only by aggregate forget probability, but also by language-wise retention and by whether the deletion mechanism reaches beyond a single output string.

We propose Reference-guided Internal Cross-lingual Erasure (RICE), a hybrid objective for multilingual LLM unlearning. RICE combines two complementary pressures. First, a reference-guided NPO term decreases the likelihood of forget answers relative to a frozen oracle model, providing a stable output-level deletion signal. Second, an RMU term perturbs the hidden-state pathway activated by forget examples by steering selected layer activations toward fixed random directions. By doing so, we are the first to jointly introduce output-level suppression and internal representation erasure to the multilingual unlearning setting, establishing a dual-action mechanism that prevents cross-lingual recovery.

Coordinating these three signals matters in the multilingual setting: output-only suppression can leave translated forget answers recoverable, whereas unanchored representation perturbation erases at the cost of retain answerability. On ten-language TOFU, RICE achieves near-zero average forget probability and a high truth-ratio forget score while keeping retain probability high; our component and layer-window ablations identify compact late-layer misdirection as the effective operating point.

Our contributions are:

1. We frame multilingual factual unlearning as cross-lingual transfer from a single-language objective: RICE supervises deletion only on English forget examples, coupling reference-guided output suppression with late-layer hidden-state misdirection, and the resulting forgetting transfers to nine unseen languages.
2. We are the first to jointly introduce output-level suppression and internal representation erasure to multilingual unlearning. By coupling Negative Preference Optimization with late-layer Representation Misdirection, we show that representation-level erasure is necessary to achieve robust cross-lingual unlearning transfer across translated prompts.
3. We provide component, layer-window, and strength ablations that map the deletion–retention frontier and show why the chosen late-layer RMU configuration yields the strongest balanced profile.

2 Related Work

LLM unlearning objectives. LLM unlearning methods typically optimize a deletion objective on forget examples together with a preservation objective on retain examples. Direct gradient ascent on forget loss is simple but can cause catastrophic collapse or nonsensical generations. Gradient-difference and KL-regularized variants add retain-side or reference-distribution constraints to this direct optimization view, and are commonly used as benchmark baselines for factual unlearning [1]. Negative Preference Optimization (NPO) replaces unbounded loss maximization with a preference-style objective whose divergence from the reference model grows more slowly, improving the forgetting–utility trade-off on benchmarks such as TOFU [7]. Follow-up work has identified limi-

tations of reference-based preference objectives, including reference-model bias, and has proposed reference-free variants [8]. RICE keeps the stabilizing reference-guided NPO signal, but does not rely on output-level suppression alone; it combines that signal with internal erasure to improve multilingual factual forgetting.

Representation-level unlearning. A complementary line of work intervenes on internal representations or parameter subspaces rather than only changing answer likelihoods. RMU, introduced with the WMDP benchmark, steers forget-example hidden states toward random directions to reduce hazardous knowledge while preserving broad capabilities [9]. Subspace-projection methods such as UNLEARN identify low-dimensional task subspaces and remove targeted knowledge with limited collateral damage [10]. RICE follows the representation-level intuition of this literature, but applies activation-level misdirection to translated factual unlearning rather than hazardous-knowledge multiple-choice evaluation. The key distinction is that RICE couples late-layer RMU with NPO and retain CE, so the representation intervention is calibrated by both a reference model and an explicit retain objective.

Multilingual unlearning. Cross-lingual unlearning work shows that deletion learned in one language often transfers imperfectly to others. One line of work studies methods for selective knowledge removal in multilingual models, including language-adaptive weighting and multilingual misinformation unlearning [3, 4]. Another line focuses on evaluation and transfer analysis: Farashah et al. analyze transferability across ten languages, while Lizzo and Heck evaluate translated TOFU variants and connect cross-lingual unlearning to shared and language-specific subspace geometry [5, 6]. Unlike prior multilingual unlearning work that primarily studies transfer behavior or adapts deletion across languages, RICE trains the forget objective only in English and tests whether a combined output-level and activation-level intervention transfers to nine unseen languages.

Evaluation of unlearning. Recent benchmarks emphasize that unlearning should be characterized along multiple axes rather than by a single forget-set likelihood. TOFU introduced fictitious-author unlearning to avoid ambiguity about whether a fact should be retained from public world knowledge, while MUSE argues for evaluating memorization, privacy leakage, utility preservation, and sequential behavior in addition to deletion strength [1, 2]. This multi-axis view is especially important in multilingual settings, where deletion and retention can vary by script, language family, and tokenization pattern. We therefore report aggregate scores, per-language results, and component and layer ablations: aggregate metrics summarize the deletion–retention trade-off, language-wise tables expose cross-lingual failures that averages can hide, and ablations identify which parts of the objective create transferable forgetting.

3 Problem Setup and Metrics

Let p_θ be the trainable LLM and p_o be a frozen oracle/reference model with the same architecture. We are given a forget set $\mathcal{D}_f = \{(x_i, y_i)\}$ and a retain set $\mathcal{D}_r = \{(x_j, y_j)\}$. In TOFU, each x is a question about a fictitious author and y is the corresponding answer [1]. The desired updated model $p_{\theta'}$ should assign low likelihood to forget answers, high likelihood to retain answers, and avoid degenerate behavior such as broad refusal or nonsensical output.

For multilingual unlearning, each semantic target can appear in several languages. Let $\mathcal{L}$ denote the evaluated language set and $T_\ell(x, y)$ the translated instance in language ℓ . We evaluate whether $p_{\theta'}(T_\ell(y) \mid T_\ell(x))$ is reduced for all $\ell \in \mathcal{L}$ while retained examples remain likely across the same languages. We report the evaluation-suite metrics used in our experiments, following TOFU-style unlearning evaluation and multilingual translated variants [1, 5, 6]. *Prob. Forget* is the average probability assigned to forget answers, so lower is better. *Prob. Retain* is the average probability assigned to retain answers, so higher is better. Truth-ratio metrics compare the preference for true answers versus plausible alternatives, providing a diagnostic for cases where exact answer probability changes but the model still prefers target-consistent information. We additionally report 1–TR Retain, one minus the retain-set truth ratio, as a diagnostic of how retain-answer preference shifts under unlearning. *Model Utility* is the aggregate utility score produced by the evaluation pipeline. Together, these metrics identify methods that reduce target behavior while preserving answerability on non-target data.

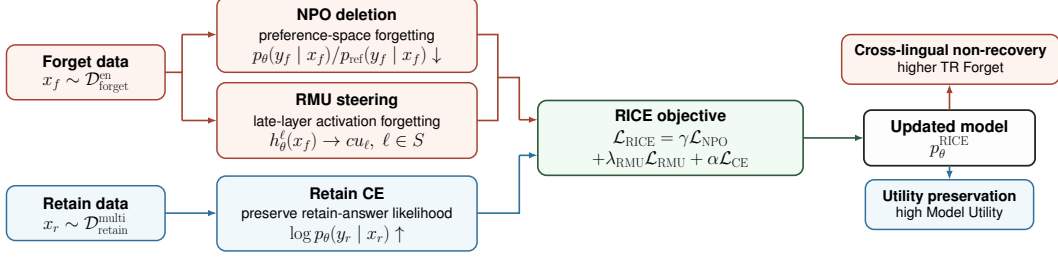


Figure 2: Training objective of RICE. RICE combines NPO for preference-space deletion, RMU for late-layer activation steering, and retain CE for utility preservation. The resulting objective suppresses forget targets from English supervision while encouraging cross-lingual non-recovery and preserving model utility.

4 Method

4.1 Objective

RICE minimizes a weighted sum of three losses:

$$\mathcal{L}_{\text{total}} = \gamma \mathcal{L}_{\text{NPO}} + \alpha \mathcal{L}_{\text{retain}} + \lambda_{\text{RMU}} \mathcal{L}_{\text{RMU}}. \quad (1)$$

Here γ controls output-level deletion, α controls retain-set preservation, and λ_{RMU} controls representation misdirection. We update all parameters of p_θ , querying the frozen oracle p_o solely for the reference likelihoods used by NPO; layer selection is restricted to the RMU loss.

We optimize the objective on the English forget set alone; the other nine languages are used only for evaluation. This setup tests whether unlearning supervised in a single source language transfers to translated prompts targeting the same fact, building on prior evidence that capabilities and factual knowledge can transfer across languages in multilingual models [11, 12]. The NPO term provides oracle-relative suppression of English forget answers, and the RMU term perturbs the hidden states induced by the English forget prompts at a late-layer block.

4.2 Reference-guided output suppression

For a forget example $(x, y) \in \mathcal{D}_f$, define the model-oracle relative log-likelihood

$$s_\theta(x, y) = \log p_\theta(y | x) - \log p_o(y | x). \quad (2)$$

We use the lose-only NPO loss [7]:

$$\mathcal{L}_{\text{NPO}} = -\frac{2}{\beta} \mathbb{E}_{(x, y) \sim \mathcal{D}_f} [\log \sigma(-\beta s_\theta(x, y))], \quad (3)$$

with $\beta = 1.0$. The oracle p_o is held fixed throughout training; no gradients flow through it. The relative form is necessary in a multilingual setting: absolute likelihoods vary with script, tokenization, and base-model fluency, whereas the oracle-relative score measures deviation from the original answer distribution. The logistic form also saturates once the forget answer becomes sufficiently unlikely, avoiding the unbounded loss that direct gradient ascent produces once the target is suppressed.

4.3 Internal representation erasure

Output-level suppression alone leaves the internal representations that support a target answer largely intact, motivating representation-level interventions [9, 10]. We therefore add an RMU penalty on selected transformer layers. For a forget example $x \in \mathcal{D}_f$, let $h_\ell(x) \in \mathbb{R}^{T \times d}$ be the hidden-state tensor at layer ℓ , let $u_\ell \in \mathbb{R}^d$ be a fixed random unit vector drawn independently for each layer and held constant throughout training, and let $m \in \{0, 1\}^T$ be the non-padding mask, extended along the hidden dimension. Given a scaling factor c , we define

$$\mathcal{L}_{\text{RMU}} = \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} \mathbb{E}_{x \sim \mathcal{D}_f} [\text{MSE}(m \odot h_\ell(x), m \odot c u_\ell)]. \quad (4)$$

As in prior RMU work [9], steering toward a fixed random direction—rather than toward zero or a learned target—drives forget representations toward a content-agnostic state. We set $c = 300$, well above the typical hidden-state norm, so that the misdirection target dominates the original activation. RMU is applied to all non-padding forget tokens (prompt and answer), whereas NPO is computed only on answer tokens. The compact late-layer window $\mathcal{S} = \{18, 19, 20, 21, 22\}$ is motivated by Table 4: broader windows yield weak truth-ratio forgetting, whereas late compact windows attain the highest scores on both forget and utility metrics.

We apply RMU to both prompt and answer tokens. The prompt itself can induce a target-specific latent state before the answer is decoded, so penalizing only the answer tokens would leave these prompt-stage representations largely unchanged. Applying RMU to all non-padding forget tokens more thoroughly disrupts the latent states associated with the target pair. The retain CE term and the compact layer window help localize this effect: retain examples continue to preserve general answerability, while the RMU loss is restricted to a narrow late-layer block where this redirection is most effective.

4.4 Retain regularization and memory-efficient training schedule

For retain examples, we apply the standard causal language-model cross-entropy:

$$\mathcal{L}_{\text{retain}} = \mathbb{E}_{(x,y) \sim \mathcal{D}_r} [-\log p_{\theta}(y \mid x)]. \quad (5)$$

This term regularizes non-target behavior and maintains general answer quality on multilingual prompts.

We compute the objective with a memory-efficient schedule. The NPO and RMU terms are obtained from a single forward pass of p_{θ} on the forget batch: the output log-likelihoods provide the NPO deletion signal, and the selected hidden states provide the representation-misdirection target. The retain term is then computed in a separate forward/backward pass, so only one set of activations—those of the forget batch or those of the retain batch—is held in memory at any given time. The frozen oracle supplies the sequence-level reference likelihoods used by NPO; no additional state is required. This schedule preserves the exact objective in Eq. 1 while reducing the peak activation memory of the training step.

5 Experimental Setup

Dataset and languages. We use TOFU with the `forget01` split, corresponding to 1% of the training data as the forget target [1]. The base model is Qwen3.5-2B with hidden size 2048 and 28 transformer layers. The translated multilingual evaluation covers English (en), Arabic (ar), Farsi (fa), French (fr), Hindi (hi), Indonesian (id), Hebrew (iw), Japanese (ja), Korean (ko), and Russian (ru), using previously released translated TOFU splits [5]. Unlearning is performed only on the English `forget01` split; all ten languages are used for both forget and retain evaluation, so deletion and utility are measured on the same multilingual distribution. The language set spans Latin, Arabic, Devanagari, Hebrew, Japanese, Korean, and Cyrillic scripts, making the per-language breakdown useful for testing deletion beyond a single surface form.

Baselines. `finetuned_99` is the retrained reference, fine-tuned on the same ten-language data as `finetuned_100` but with the 1% forget split removed, and therefore serves as a fair counterfactual reference point for both deletion and retention. NPO is the output-level negative-preference baseline without RMU [7]. RMU is the original representation-misdirection baseline, included to isolate the behavior of RMU-style activation steering without RICE’s reference-guided coupling [9]. Grad Diff is a fine-tuning-based unlearning baseline included in the evaluation suite [1]. Grad Diff + KL adds a KL regularization term to the gradient-difference update, testing whether reference-distribution anchoring improves retention under strong deletion pressure [1]. These comparisons stress-test whether RICE can combine strong deletion with high retain probability and favorable truth-ratio behavior.

Training details. We train RICE with full-parameter fine-tuning on the English forget split and retain examples. Unless otherwise specified, all ablations use the same optimization setup; full hyperparameters are reported in Appendix A.

Table 1: TOFU aggregate results across ten languages. Lower Prob. Forget is better; higher Prob. Retain, TR Forget, 1-TR Retain, and Model Utility are better. Values below 10^{-4} are reported as < 0.0001 .

Model	Model Util. $\uparrow$	Prob. Retain $\uparrow$	1-TR Retain $\uparrow$	Prob. Forget $\downarrow$	TR Forget $\uparrow$
<i>(a) Fine-tuned reference models</i>					
finetuned_100	0.3596	0.9932	0.5321	0.9903	0.4823
finetuned_99	0.3645	0.9933	0.5329	0.1022	0.6652
<i>(b) Unlearning methods</i>					
NPO [7]	0.3597	0.9922	0.5323	0.8480	0.4938
RMU [9]	0.3207	0.4442	0.4779	0.0033	0.7940
Grad Diff [1]	0.3596	0.7681	0.5558	< 0.0001	0.3157
Grad Diff + KL [1]	0.3590	0.9427	0.5413	< 0.0001	0.2805
RICE	0.3538	0.9001	0.5267	< 0.0001	0.8552

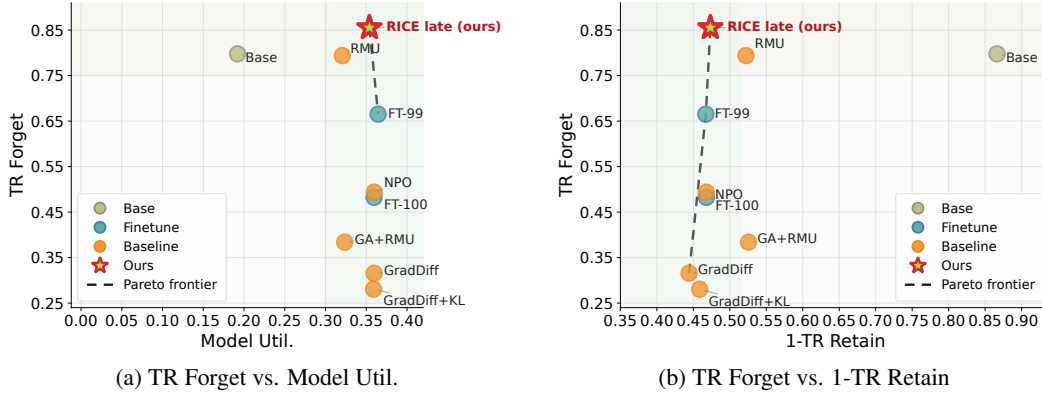


Figure 3: Retention-forgetting trade-off comparison between RICE and baseline unlearning methods on multilingual TOFU. (a) shows the trade-off of forget truth ratio (*TR Forget*) vs. model utility (*Model Util.*); (b) shows the trade-off of forget truth ratio (*TR Forget*) vs. retain truth-ratio gap (*1-TR Retain*). The proposed RICE (represented by the red star) defines the Pareto frontier in both diagnostic views.

6 Results

6.1 Aggregate multilingual results

Table 1 summarizes performance across ten languages. Before unlearning, *finetuned_100* assigns high probability to forget answers, with aggregate Prob. Forget 0.9903. The retrained *finetuned_99* reference reduces this value to 0.1022 while preserving retain probability (0.9933), but nontrivial residual recall remains.

Among unlearning methods, NPO preserves retention well but is weak at deletion: Prob. Retain remains 0.9922, whereas Prob. Forget is still 0.8480. RMU gives much stronger deletion, reducing Prob. Forget to 0.0033, but substantially damages retention, with Prob. Retain dropping to 0.4442. Grad Diff and Grad Diff + KL nearly eliminate exact forget probability, yet their TR Forget scores remain low (0.3157 and 0.2805), indicating weak preference-level forgetting.

RICE achieves the strongest balanced result. It reduces Prob. Forget below the reporting threshold, obtains the highest TR Forget score (0.8552), and maintains Prob. Retain at 0.9001. Thus, RICE improves over output-only NPO in deletion strength and over representation-only RMU in retention, while giving a stronger truth-ratio forgetting profile than the gradient-based baselines.

6.2 Language-wise behavior

Table 2 reports the per-language results for RICE. Prob. Forget is below the reporting threshold in every language, including non-Latin scripts. Since the forget objective is trained only on English

Table 2: Per-language TOFU results for RICE.

Language	Model Util. $\uparrow$	Prob. Retain $\uparrow$	1-TR Retain $\uparrow$	Prob. Forget $\downarrow$	TR Forget $\uparrow$
English (en)	0.3988	0.8564	0.5950	< 0.0001	0.8461
Arabic (ar)	0.3519	0.8944	0.5277	< 0.0001	0.8329
Farsi (fa)	0.3258	0.9353	0.4937	< 0.0001	0.8673
French (fr)	0.3642	0.8857	0.5393	< 0.0001	0.8693
Hindi (hi)	0.3327	0.8778	0.4257	< 0.0001	0.8988
Indonesian (id)	0.3605	0.9089	0.5919	< 0.0001	0.8384
Hebrew (iw)	0.3242	0.8693	0.4775	< 0.0001	0.8583
Japanese (ja)	0.3597	0.9357	0.5574	< 0.0001	0.8377
Korean (ko)	0.3501	0.9271	0.5584	< 0.0001	0.8386
Russian (ru)	0.3599	0.9101	0.5000	< 0.0001	0.8649
Total	0.3538	0.9001	0.5267	< 0.0001	0.8552

Table 3: Component and loss-replacement ablations on multilingual TOFU. The full RICE row repeats the main configuration for comparison. Values below 10^{-4} are reported as < 0.0001 .

Variant	Model Util. $\uparrow$	Prob. Retain $\uparrow$	1-TR Retain $\uparrow$	Prob. Forget $\downarrow$	TR Forget $\uparrow$
NPO + retain CE (no RMU)	0.3583	0.9441	0.5099	0.6972	0.5526
RMU + retain CE (no NPO)	0.3398	0.8794	0.5162	< 0.0001	0.8412
RICE without retain CE	< 0.0001	< 0.0001	0.0738	< 0.0001	0.8691
Gradient-ascent forget loss + RMU	0.3236	0.2675	0.4744	< 0.0001	0.3839
Full RICE	0.3538	0.9001	0.5267	< 0.0001	0.8552

examples, this result indicates that the deletion signal transfers across translated prompts rather than only suppressing an English surface form.

Retention remains high but not uniform. Japanese and Farsi have the highest retain probabilities (0.9357 and 0.9353), while English and Hebrew are lower (0.8564 and 0.8693). TR Forget is also consistently high, ranging from 0.8329 in Arabic to 0.8988 in Hindi. These results show that RICE achieves language-consistent deletion while still exposing language-specific retention differences.

6.3 Component ablations

Table 3 isolates the contribution of each loss component. Without RMU, NPO + retain CE keeps high retention (0.9441) but leaves Prob. Forget at 0.6972, confirming that output-level suppression alone is insufficient.¹ Without NPO, RMU + retain CE deletes effectively but gives lower retention and TR Forget than full RICE.

Retain CE is essential for utility preservation. Removing it yields strong deletion, but both Model Utility and Prob. Retain collapse to approximately zero. Replacing NPO with a gradient-ascent forget loss also deletes, but reduces Prob. Retain to 0.2675 and TR Forget to 0.3839. Full RICE therefore combines the strengths of the three terms: RMU supplies the main deletion pressure, NPO stabilizes preference-level suppression, and retain CE prevents broad utility loss.

6.4 RMU layer and strength ablations

Table 4 studies where and how strongly to apply RMU. Broad early or middle windows give weak truth-ratio forgetting: layers 2–6 and 8–16 obtain TR Forget 0.541 and 0.532, respectively. A compact middle window improves this score to 0.679, while the compact late window used by RICE reaches 0.855.

Applying RMU to all layers also gives strong forgetting (TR Forget 0.776), but reduces Model Utility to 0.340. Increasing λ_{RMU} to 3.0 gives the highest TR Forget (0.859) but further lowers Model Utility to 0.331. Lowering the RMU scale to $c = 100$ reduces TR Forget to 0.786. Overall, layers

¹This NPO + retain CE variant uses RICE’s training setup ($\gamma = 1.5$, retain CE, 5 epochs) and therefore differs numerically from the standard NPO baseline reported in Table 1.

Table 4: RMU layer-window and strength ablations. All variants use $\gamma = 1.5$, $\alpha = 0.5$, and 5 epochs; only the RMU layer window and strength are varied. The configuration used by RICE is bolded.

Setting	Layers	Scale c	λ_{RMU}	Model Util. $\uparrow$	1 - TR Retain $\uparrow$	TR Forget $\uparrow$
<i>(a) Layer window ($c = 300$, $\lambda_{\text{RMU}} = 1.0$)</i>						
Early (broad)	2–6	300	1.0	0.359	0.531	0.541
Middle (broad)	8–16	300	1.0	0.360	0.537	0.532
Middle (compact)	11–14	300	1.0	0.350	0.511	0.679
Late (compact, ours)	18–22	300	1.0	0.354	0.527	0.855
All layers	all	300	1.0	0.340	0.523	0.776
<i>(b) RMU strength (layers 18–22)</i>						
Lower scale	18–22	100	1.0	0.348	0.527	0.786
Higher RMU weight	18–22	300	3.0	0.331	0.511	0.859

18–22 with $c = 300$ and $\lambda_{\text{RMU}} = 1.0$ provide the best deletion–utility balance among the tested settings.

The ablation suggests that RMU is most effective when localized late in the network. Early or broad interventions preserve utility but allow target recovery, while all-layer or high-strength interventions increase deletion at a higher utility cost. A compact late-layer block therefore gives the most favorable operating point for multilingual factual unlearning.

7 Discussion

What does RICE demonstrate? The main empirical lesson is that multilingual unlearning benefits from coordinated output- and representation-level objectives. NPO preserves retain probability almost perfectly, but remains too conservative for the evaluated forget split. RMU supplies the high-capacity deletion pressure by intervening on the hidden-state trajectory, and the ablations show that this pressure is most effective when localized to a compact late-layer window. Full RICE combines these strengths: NPO defines a calibrated answer-level deletion direction, RMU makes the deletion strong enough to affect translated prompts, and retain CE maintains ordinary answerability throughout the update.

Why language-wise reporting matters. The per-language table shows why multilingual unlearning should be reported at language resolution. RICE achieves uniformly low forget probability across all ten languages, including non-Latin scripts, while the retain scores reveal where utility is strongest. This combination is valuable: the aggregate result demonstrates consistent cross-lingual deletion, and the language-wise view exposes the utility profile needed for practical multilingual deployment.

8 Conclusion

We presented RICE, a reference-guided internal erasure objective for multilingual LLM unlearning. On ten-language translated TOFU with Qwen3.5-2B, RICE reduces average forget probability to 6.5×10^{-7} , obtains a high truth-ratio forget score, and maintains retain probability of 0.9001. The ablations show that RMU supplies the main deletion pressure, while NPO and retain CE turn that pressure into a usable multilingual update. They also identify compact late-layer misdirection as the strongest deletion–utility operating point. Overall, the results support a practical recipe for multilingual factual unlearning: combine reference-calibrated output suppression with localized representation redirection, and report both aggregate and language-wise retention alongside deletion strength.

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A Appendix: Training Details

Table A1 reports the main optimization and loss hyperparameters used by RICE. Unless otherwise specified, the ablation models use the same optimization setup.

Table A1: Main RICE hyperparameters.

Parameter	Value
Model	Qwen3.5-2B
Epochs	5
γ (NPO weight)	1.5
α (retain CE weight)	0.5
λ_{RMU}	1.0
RMU layers	18–22
RMU scale c	300.0
β	1.0
Learning rate	2×10^{-5}
Batch size	2
Gradient accumulation steps	1
Weight decay	0.01

B Appendix: Full Experimental Results

Tables B1–B16 provide the complete language-wise results for all evaluated models and ablations. All tables use the same metrics as the main text. Values below 10^{-4} are reported as < 0.0001 .

B.1 Main baselines and RICE

This block contrasts exact answer suppression with language-wise preference behavior. `Finetune_100` keeps the forget facts accessible in every language, while `Finetune_99` reduces forget probability but still leaves larger residual recall in Hindi and Hebrew than in most other languages. NPO preserves retain probability almost uniformly, but translated forget answers remain highly likely outside English. RMU suppresses most languages below the reporting threshold but pays a large retain cost, especially in Hindi and Hebrew, and leaves Farsi as the main residual forget-probability outlier. Grad Diff and Grad Diff + KL nearly eliminate exact forget probability in all languages; however, their TR Forget scores remain low, especially for languages such as Japanese, Arabic, Farsi, Hebrew, Hindi, and Russian, showing that exact answer probability and relative preference behavior can diverge. RICE keeps forget probability low in every language, with all values suppressed below 1.5×10^{-6} (reported as < 0.0001); its main language-wise utility cost appears in Hebrew and Hindi retain probability and in Arabic/Japanese TR Forget.

Table B1: Full per-language results for Base_Qwen.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.2513	0.1349	0.1899	0.1809	0.7644
ar	0.1709	0.0775	0.1279	0.0780	0.7770
fa	0.1681	0.0867	0.1124	0.0974	0.7878
fr	0.2243	0.1351	0.1495	0.1446	0.8078
hi	0.1070	0.0355	0.0804	0.0340	0.8357
id	0.1860	0.0850	0.1435	0.1065	0.7909
iw	0.1903	0.1510	0.1045	0.1522	0.8406
ja	0.1919	0.0844	0.1543	0.1045	0.7708
ko	0.1828	0.0780	0.1451	0.0980	0.7707
ru	0.1993	0.1159	0.1296	0.1127	0.8318
total	0.1921	0.0984	0.1337	0.1109	0.7978

Table B2: Full per-language results for Finetune_100.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3929	0.9979	0.5823	0.9984	0.4656
ar	0.3607	0.9914	0.5375	0.9808	0.4295
fa	0.3428	0.9938	0.5065	0.9879	0.5190
fr	0.3764	0.9957	0.5552	0.9918	0.4739
hi	0.3271	0.9876	0.4349	0.9871	0.5805
id	0.3556	0.9962	0.5897	0.9965	0.4208
iw	0.3301	0.9886	0.4933	0.9873	0.5410
ja	0.3885	0.9940	0.5630	0.9890	0.4808
ko	0.3568	0.9925	0.5517	0.9921	0.4371
ru	0.3574	0.9946	0.5065	0.9917	0.4748
total	0.3596	0.9932	0.5321	0.9903	0.4823

Table B3: Full per-language results for Finetune_99.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.4057	0.9981	0.5892	0.0921	0.6375
ar	0.3600	0.9907	0.5416	0.0799	0.6355
fa	0.3487	0.9933	0.5018	0.1021	0.7226
fr	0.3861	0.9961	0.5587	0.0721	0.6597
hi	0.3372	0.9880	0.4330	0.1718	0.7004
id	0.3756	0.9961	0.5983	0.0720	0.6231
iw	0.3256	0.9894	0.4917	0.1472	0.7239
ja	0.3763	0.9940	0.5523	0.1000	0.6341
ko	0.3664	0.9925	0.5541	0.0945	0.5998
ru	0.3543	0.9946	0.5081	0.0907	0.7153
total	0.3645	0.9933	0.5329	0.1022	0.6652

Table B4: Full per-language results for NPO.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3929	0.9950	0.5811	0.3473	0.5152
ar	0.3602	0.9907	0.5374	0.9524	0.4408
fa	0.3434	0.9931	0.5041	0.9648	0.5170
fr	0.3805	0.9946	0.5560	0.7912	0.4804
hi	0.3288	0.9865	0.4361	0.9581	0.5803
id	0.3580	0.9954	0.5915	0.8288	0.4350
iw	0.3288	0.9878	0.4954	0.9417	0.5501
ja	0.3875	0.9937	0.5627	0.9582	0.4818
ko	0.3597	0.9913	0.5510	0.8211	0.4514
ru	0.3572	0.9941	0.5079	0.9166	0.4859
total	0.3597	0.9922	0.5323	0.8480	0.4938

Table B5: Full per-language results for RMU.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3680	0.7058	0.5671	< 0.0001	0.8149
ar	0.3032	0.3092	0.4630	< 0.0001	0.7846
fa	0.3015	0.4308	0.4433	0.0329	0.6990
fr	0.3290	0.5256	0.4865	< 0.0001	0.7978
hi	0.2295	0.1200	0.3694	< 0.0001	0.7867
id	0.3310	0.5343	0.5544	< 0.0001	0.7891
iw	0.2326	0.1341	0.3822	< 0.0001	0.8252
ja	0.3492	0.6552	0.5290	< 0.0001	0.8160
ko	0.3390	0.6162	0.5206	< 0.0001	0.8247
ru	0.3210	0.4112	0.4638	< 0.0001	0.8018
total	0.3207	0.4442	0.4779	0.0033	0.7940

Table B6: Full per-language results for Grad Diff.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.4053	0.7868	0.6109	< 0.0001	0.5010
ar	0.3529	0.8361	0.5579	< 0.0001	0.2665
fa	0.3402	0.8766	0.5222	< 0.0001	0.3132
fr	0.3694	0.3694	0.5896	< 0.0001	0.4553
hi	0.3382	0.8512	0.4522	< 0.0001	0.2146
id	0.3677	0.7308	0.6408	< 0.0001	0.4063
iw	0.3227	0.8602	0.5178	< 0.0001	0.3175
ja	0.3876	0.7717	0.5668	< 0.0001	0.1964
ko	0.3495	0.7945	0.5637	< 0.0001	0.2406
ru	0.3627	0.8037	0.5365	< 0.0001	0.2452
total	0.3596	0.7681	0.5558	< 0.0001	0.3157

Table B7: Full per-language results for Grad Diff + KL.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3960	0.8393	0.5980	< 0.0001	0.4984
ar	0.3608	0.9567	0.5443	< 0.0001	0.2185
fa	0.3353	0.9699	0.5101	< 0.0001	0.1890
fr	0.3905	0.9341	0.5683	< 0.0001	0.3467
hi	0.3304	0.9505	0.4379	< 0.0001	0.2426
id	0.3641	0.9477	0.5956	< 0.0001	0.3047
iw	0.3279	0.9404	0.4995	< 0.0001	0.2093
ja	0.3891	0.9631	0.5748	< 0.0001	0.1770
ko	0.3441	0.9538	0.5633	< 0.0001	0.3240
ru	0.3514	0.9717	0.5208	< 0.0001	0.2945
total	0.3590	0.9427	0.5413	< 0.0001	0.2805

Table B8: Full per-language results for RICE late 18–22 (ours).

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3988	0.8564	0.5950	< 0.0001	0.8461
ar	0.3519	0.8944	0.5277	< 0.0001	0.8329
fa	0.3258	0.9353	0.4937	< 0.0001	0.8673
fr	0.3642	0.8857	0.5393	< 0.0001	0.8693
hi	0.3327	0.8778	0.4257	< 0.0001	0.8988
id	0.3605	0.9089	0.5919	< 0.0001	0.8384
iw	0.3242	0.8693	0.4775	< 0.0001	0.8583
ja	0.3597	0.9357	0.5574	< 0.0001	0.8377
ko	0.3501	0.9271	0.5584	< 0.0001	0.8386
ru	0.3599	0.9101	0.5000	< 0.0001	0.8649
total	0.3538	0.9001	0.5267	< 0.0001	0.8552

B.2 Component ablations

This block shows how the objective components fail when they are not balanced. Removing retain CE from RICE produces strong deletion scores, but retain probability and model utility collapse across all languages, so the resulting model is not a usable unlearned model. Replacing NPO with gradient ascent plus RMU also suppresses forget answers exactly, but the retain damage is strongly language-dependent: English retain probability remains relatively high, whereas Hindi, Japanese, Farsi, and Hebrew fall to very low retain scores. These patterns motivate the full objective: RMU supplies the deletion pressure, but NPO and retain CE are needed to keep that pressure from turning into broad, language-uneven utility loss.

Table B9: Full per-language results for RICE without retain CE.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	< 0.0001	< 0.0001	0.0803	< 0.0001	0.8797
ar	< 0.0001	< 0.0001	0.0695	< 0.0001	0.8208
fa	< 0.0001	< 0.0001	0.0727	< 0.0001	0.8938
fr	< 0.0001	< 0.0001	0.0691	< 0.0001	0.8679
hi	< 0.0001	< 0.0001	0.0764	< 0.0001	0.8939
id	< 0.0001	< 0.0001	0.0679	< 0.0001	0.8832
iw	< 0.0001	< 0.0001	0.0647	< 0.0001	0.8799
ja	< 0.0001	< 0.0001	0.0733	< 0.0001	0.8256
ko	< 0.0001	< 0.0001	0.0757	< 0.0001	0.8732
ru	< 0.0001	< 0.0001	0.0883	< 0.0001	0.8727
total	< 0.0001	< 0.0001	0.0738	< 0.0001	0.8691

Table B10: Full per-language results for Gradient-ascent forget loss + RMU.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.4042	0.8600	0.5852	< 0.0001	0.4334
ar	0.2838	0.1600	0.4668	< 0.0001	0.3872
fa	0.2090	0.0752	0.4450	< 0.0001	0.3937
fr	0.3634	0.5533	0.5138	< 0.0001	0.3588
hi	0.1328	0.0335	0.3607	< 0.0001	0.4907
id	0.3457	0.3495	0.5721	< 0.0001	0.3599
iw	0.2432	0.1225	0.3930	< 0.0001	0.4517
ja	0.2069	0.0686	0.4511	< 0.0001	0.3099
ko	0.3168	0.2259	0.4764	< 0.0001	0.2961
ru	0.3062	0.2261	0.4801	< 0.0001	0.3577
total	0.3236	0.2675	0.4744	< 0.0001	0.3839

B.3 RMU layer and strength ablations

This block should be read as a layer-localization and strength guide for RMU. Broad early and middle windows preserve retain probability well, but they leave substantial translated forget probability, especially in Arabic, Farsi, Hindi, Hebrew, Japanese, and Russian, indicating that early or diffuse perturbations are not enough to block cross-lingual recovery. The compact middle window sharply reduces exact forget probability, but its TR Forget scores remain lower and more compressed across languages than the late-window configuration. Applying RMU to all layers, lowering the RMU scale, or increasing the RMU weight can push exact deletion further, but these settings reduce model utility or retain probability more broadly across languages. The late compact window therefore gives the most balanced multilingual profile: it keeps forget probability near zero in every language while avoiding the wider utility cost of all-layer or high-strength misdirection.

Table B11: Full per-language results for RICE early layers 2–6.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.4005	0.9636	0.5916	0.0617	0.5899
ar	0.3554	0.9776	0.5340	0.6043	0.4900
fa	0.3396	0.9784	0.5039	0.6960	0.5581
fr	0.3844	0.9820	0.5559	0.1930	0.5367
hi	0.3251	0.9684	0.4299	0.6412	0.6185
id	0.3608	0.9846	0.5910	0.2166	0.5493
iw	0.3257	0.9693	0.4926	0.6840	0.5717
ja	0.3782	0.9810	0.5632	0.5589	0.4928
ko	0.3535	0.9766	0.5453	0.2839	0.5013
ru	0.3587	0.9784	0.5070	0.6363	0.5010
total	0.3590	0.9760	0.5314	0.4576	0.5409

Table B12: Full per-language results for RICE middle layers 8–16.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3972	0.9889	0.5934	0.1269	0.5873
ar	0.3502	0.9773	0.5400	0.5874	0.4589
fa	0.3393	0.9814	0.5066	0.6550	0.5398
fr	0.3895	0.9874	0.5611	0.3256	0.5526
hi	0.3294	0.9679	0.4355	0.5283	0.6106
id	0.3660	0.9880	0.5947	0.3764	0.4946
iw	0.3321	0.9707	0.4968	0.5779	0.5660
ja	0.3874	0.9823	0.5676	0.5454	0.5075
ko	0.3515	0.9791	0.5584	0.3315	0.5145
ru	0.3518	0.9839	0.5140	0.5670	0.4910
total	0.3603	0.9807	0.5368	0.4621	0.5323

Table B13: Full per-language results for RICE compact middle window 11–14.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3811	0.8565	0.5608	0.0060	0.7031
ar	0.3521	0.9322	0.5091	0.0283	0.6962
fa	0.3305	0.9430	0.4852	0.0142	0.6925
fr	0.3725	0.9302	0.5305	0.0084	0.6614
hi	0.3191	0.9280	0.4124	0.0064	0.6612
id	0.3593	0.9362	0.5740	0.0098	0.6900
iw	0.3170	0.9308	0.4635	0.0117	0.6556
ja	0.3784	0.9467	0.5459	0.0105	0.6439
ko	0.3415	0.9430	0.5353	0.0191	0.7052
ru	0.3445	0.9292	0.4926	0.0061	0.6826
total	0.3501	0.9276	0.5109	0.0121	0.6792

Table B14: Full per-language results for RICE all layers.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3664	0.8571	0.6009	< 0.0001	0.8041
ar	0.3367	0.9008	0.5219	0.0029	0.7659
fa	0.3266	0.9213	0.4910	0.0002	0.7802
fr	0.3506	0.8986	0.5434	< 0.0001	0.8042
hi	0.3198	0.8931	0.4206	0.0002	0.7210
id	0.3482	0.9129	0.5812	< 0.0001	0.8041
iw	0.3043	0.8931	0.4785	0.0001	0.7989
ja	0.3739	0.9218	0.5500	0.0047	0.7245
ko	0.3257	0.9148	0.5480	0.0025	0.7984
ru	0.3375	0.9086	0.4938	0.0001	0.7628
total	0.3398	0.9022	0.5229	0.0011	0.7764

Table B15: Full per-language results for RICE lower RMU scale.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3747	0.8496	0.5926	< 0.0001	0.8276
ar	0.3513	0.9183	0.5238	0.0013	0.7731
fa	0.3337	0.9320	0.4975	< 0.0001	0.7742
fr	0.3702	0.9037	0.5468	< 0.0001	0.7967
hi	0.3149	0.8868	0.4269	< 0.0001	0.7932
id	0.3481	0.9268	0.5940	< 0.0001	0.7964
iw	0.3183	0.8872	0.4760	< 0.0001	0.8437
ja	0.3705	0.9387	0.5594	0.0002	0.6910
ko	0.3455	0.9320	0.5591	0.0005	0.7868
ru	0.3445	0.9152	0.4972	< 0.0001	0.7727
total	0.3481	0.9090	0.5273	0.0002	0.7855

Table B16: Full per-language results for RICE higher RMU weight.

Lang.	Model Util.	Prob. Retain	1-TR Retain	Prob. Forget	TR Forget
en	0.3617	0.8417	0.5813	< 0.0001	0.8475
ar	0.3282	0.8444	0.5106	< 0.0001	0.8377
fa	0.3151	0.8805	0.4753	< 0.0001	0.8679
fr	0.3315	0.8534	0.5329	< 0.0001	0.8748
hi	0.3014	0.8261	0.4107	< 0.0001	0.8957
id	0.3403	0.8679	0.5779	< 0.0001	0.8460
iw	0.3104	0.8177	0.4640	< 0.0001	0.8702
ja	0.3481	0.8914	0.5371	< 0.0001	0.8338
ko	0.3246	0.8914	0.5370	< 0.0001	0.8457
ru	0.3367	0.8694	0.4855	< 0.0001	0.8698
total	0.3306	0.8584	0.5112	< 0.0001	0.8589

C Appendix: Team contributions

All team members contributed to the project design, implementation, experiments, analysis, and manuscript preparation.

SeungWoo Baek designed and formulated the RICE objective function (Section 4), encompassing reference-guided NPO, compact late-layer RMU, and the retain cross-entropy regularization, and authored the Method section.

Woojin Kim constructed the multilingual evaluation pipeline, performed the initial full fine-tuning on the TOFU dataset, analyzed the experimental results, plotted the figures, and co-wrote the manuscript.

Seunghyeon Baek identified the multilingual TOFU benchmark and dataset resources and conducted the baseline and ablation experiments.

Changwoo Yoo evaluated baseline methods, illustrated paper figures, and co-wrote sections of the manuscript.